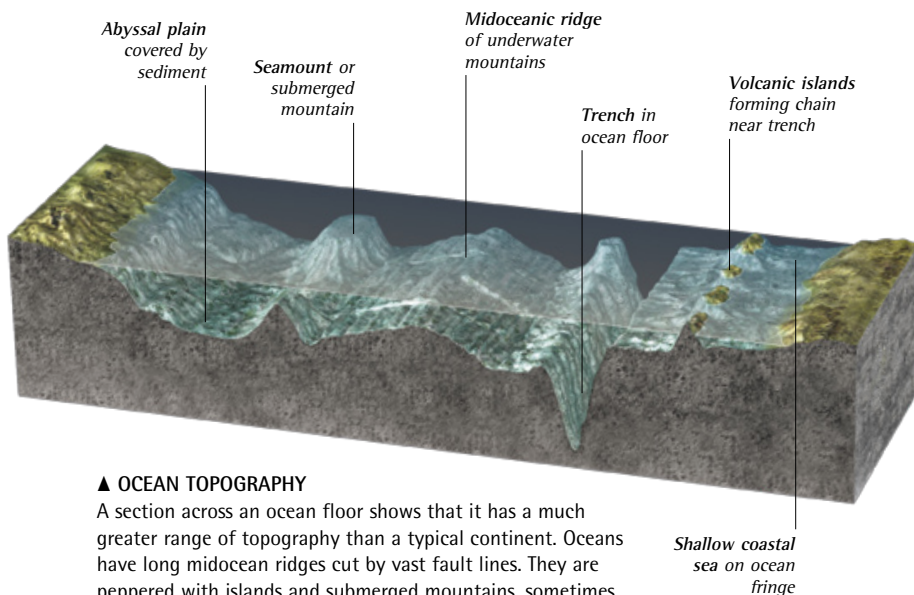


THE OCEAN FLOORS

Since the ocean depths were first investigated in the 19th century, scientists have known that they are not just featureless basins filled with water. They found that oceans are fringed by shallow coastal seas, and that the deeper regions are dotted with submarine volcanoes and long mountain ridges. They also discovered trenches that plunge to extraordinary depths in the ocean floors. The data collected enabled them to produce amazing maps showing these features in graphic detail.



▲ OCEAN TOPOGRAPHY

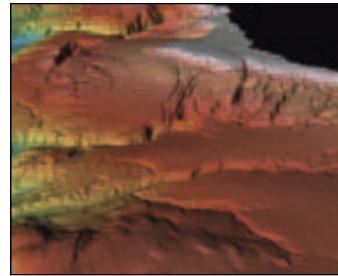
A section across an ocean floor shows that it has a much greater range of topography than a typical continent. Oceans have long mid-ocean ridges cut by vast fault lines. They are peppered with islands and submerged mountains, sometimes forming long chains. The fringes of most of the Pacific and northeast Indian Ocean are marked by deep trenches, some of which are three times the average depth of the world's oceans.

► EXAGGERATED HEIGHTS AND SLOPES

On all maps and illustrations of the ocean floor, vertical distances are hugely exaggerated so that the features are easy to see. Undersea volcanoes and islands seem to rise from the ocean floor like needles, while trenches and shelves look like canyons and cliffs. In reality, these features are far less dramatic.

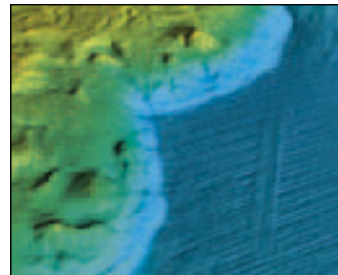
Volcanic island is shown far too high

OCEAN FLOOR FEATURES



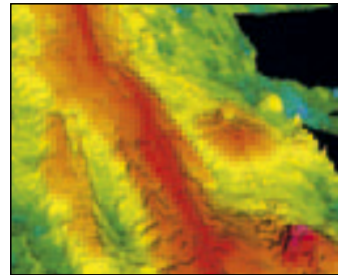
CONTINENTAL SHELVES

The shallow regions at the fringes of oceans are the continental shelves. They can extend a long way from the coast, seen here in black, and are often cut by deep canyons. At their outer edges, the continental shelves slope down to the ocean floor.



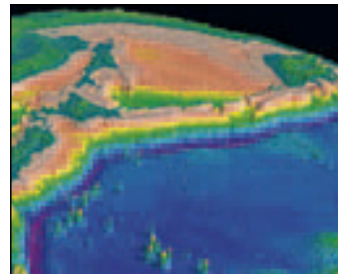
ABYSSAL PLAINS

Great tracts of ocean floor are covered with soft mud, sand, and other sediments. These form the abyssal plains, seen here in blue. The layers of sediments are often very deep, masking the rugged contours of the oceanic bedrock below.



MIDOCEAN RIDGES

Shown in red is an isolated seamount and part of a mid-ocean ridge. These ridges are the longest mountain chains on Earth, and they join up to form a network around the globe. They all form rugged peaks of dark basalt rock.



OCEAN TRENCHES

A satellite image shows an ocean trench in dark blue, lying between the deep ocean and the shallow seas off Asia. Linked with earthquakes and volcanoes, some of these trenches are deep enough to swallow the highest mountains on Earth.